

Lecture 15

Four Fundamental Subspaces

Instructors:

L01 Cosme Louart 程高木

L02 Junfeng Wu 吴均峰

L03&L04 Ruoyu Sun 孙若愚



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

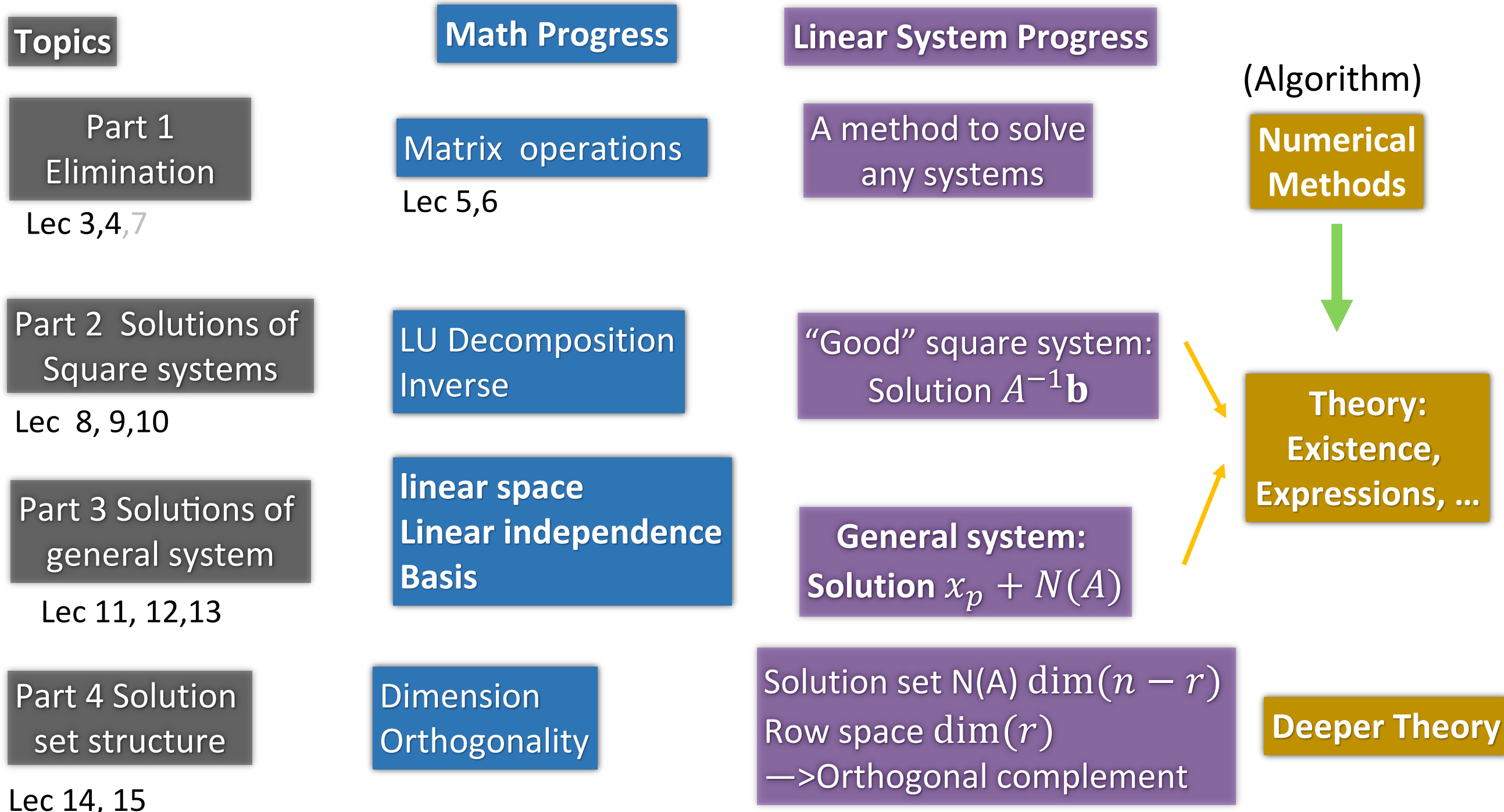
数据科学学院

School of Data Science

Roadmap: *Topics Before Midterm*

Lec 1,2: Vector operations.

Theme of Lec 3-15: Solving Linear Systems



Today's Lecture: Outline and Goals

Today ... Four Fundamental Subspaces

1. Rank and # of solutions (cont'd) Strang's book: Sec 3.4, 3.5
3. Orthogonal complement
4. Four fundamental subspaces

After this lecture, you should be able to

- Tell the # of solutions for full rank matrices
- Judge orthogonal complement of another subspace
- Tell four fundamental subspaces and their relations

Review of Related Contents

Review of Linear Independence and Basis

$\mathbf{u}_1, \dots, \mathbf{u}_n$ are **linearly independent** iff the following holds:

$c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = \mathbf{0}$ only happens when $c_1 = \dots = c_k = 0$.

$\mathbf{u}_1, \dots, \mathbf{u}_n$ form a **basis** of V iff the following two hold:

- i) they are linearly independent; [no redundant information; not too many]
- ii) they span V . [no loss of information; not too few]

Eg1 $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$. (called “standard basis”)

Eg2 $\{E_{ij}, i = 1, \dots, m; j = 1, \dots, n\}$ is a basis of $\mathbb{R}^{m \times n}$.

Review of Dimension and Rank

Theorem 14.1 (bases have same size)

If $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ and $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ are bases of a linear space V , then $m = n$.

First big
supporting
Theorem
in this course!

All bases have the same size!! We call it “dimension”!

Theorem 14.3

Row-rank and column-rank matrix are the same.

Second big
supporting
Theorem
in this course!

Supporting theorems:
They are so basic that you will
take them for granted after 1 year,
or even forget they are “theorems”!

Proposition (rank = # of pivots)

Suppose when using Gauss-Jordan elimination to solve
a linear system $Ax = 0$, there are r pivots.

Then $\text{rank}(A) = r$.

$\dim(\text{Row}(A)) = \dim(\text{C}(A)) = \# \text{ of pivots.}$

Summary: Number of Solutions v.s. Rank

Assume the pivot columns appear at the beginning.
(if not, the analysis is still similar; omit here)

$$\begin{bmatrix} I_r & F \\ \mathbf{0}_{(m-r) \times r} & \mathbf{0}_{(m-r) \times (n-r)} \end{bmatrix} \begin{bmatrix} x_P \\ x_F \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{b}}_{r \times 1} \\ \mathbf{c}_{(m-r) \times 1} \end{bmatrix}. \quad \text{Here, \# of pivots } r = \text{rank.}$$

Dependent on how large n, m, r , some of the blocks may disappear.

Four types	$[I]$	$[I \ F]$	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
Their ranks	$r = m = n$	$r = m < n$	$r = n < m$	$r < m, \quad r < n$
# of solutions	1	∞	0 or 1	0 or ∞

Group Case 1 & 2:

$r = m$
of solutions: 1 or ∞

$r = n$ (Group Case 1 & Case 3).
of solutions: 0 or 1

Part I Number of solutions v.s. Rank

Full Rank Matrix

Definition 14.4 (full rank matrix)

Let $A \in \mathbb{R}^{m \times n}$.

If $\text{rank}(A) = m$, then we say A has full row rank.

If $\text{rank}(A) = n$, then we say A has full column rank.

If $\text{rank}(A) = \min\{m, n\}$, then we say A has full rank.

Wording: Can also say “ A is a full rank matrix”; or A is of full rank.

E.g. Is this matrix of full rank, full row rank, full column rank?

$$m=4 \begin{bmatrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1} \\ 0 & 0 & 0 \end{bmatrix}$$

$n=3$

$$\text{rank}(A) = 3 = n < m$$

Full Column Rank

$n-r=0$

Proposition 15.1: If $\text{rank}(A) = n$, i.e., full column rank, then for any b , $Ax = b$ has **at most one** solution (# of solutions must be 0 or 1; **eliminate possibility of ∞**).

From RREF perspective:

No free columns in RREF $\rightarrow$ **No free variables** $\rightarrow$ impossible to have >1 solutions.

After re-ordering columns of RREF, we get:

Pivot columns free columns

pivot rows free rows

$$\begin{matrix} r \\ m-r \end{matrix} \left\{ \begin{matrix} \left[\begin{matrix} I & F \end{matrix} \right] \\ \left[\begin{matrix} 0 & 0 \end{matrix} \right] \end{matrix} \right\} \begin{bmatrix} x_p \\ x_F \end{bmatrix} = \begin{bmatrix} \hat{b} \\ c \end{bmatrix}$$

$\underbrace{\quad}_r \quad \underbrace{\quad}_{n-r}$

determine # of solutions

$$\begin{cases} x_1 = \square - \square x_2 \\ \vdots \\ \vdots \end{cases} \quad \downarrow \text{free var}$$

Full Column Rank

Proposition 15.1: If $\text{rank}(A) = n$, i.e., full column rank, then for any b , $Ax = b$ has at most one solution (# of solutions must be 0 or 1; eliminate possibility of ∞).

Proof from vector perspective:

Suppose $A = [u_1, \dots, u_n]$

vectors u_i 's are independent $\Rightarrow$ at most one way to represent any v by u_i 's.

Directly prove from vector perspective:

Suppose v can be represented in two ways:

$$v = \alpha_1 u_1 + \dots + \alpha_n u_n$$

$$v = \beta_1 u_1 + \dots + \beta_n u_n$$

Then $0 = (\alpha_1 - \beta_1)u_1 + \dots + (\alpha_n - \beta_n)u_n$ LC of u_i 's = 0.

By independence of u_i 's, we have $\alpha_1 - \beta_1 = \dots = \alpha_n - \beta_n = 0$; so $\alpha_i = \beta_i, \forall i$.

So there is at most one way to represent v by u_i 's.

Unique Representation

When v is in the span of u_i 's, "at most one way" becomes "exactly one way".

Implication: unique representation by independent set

Corollary 15.1: Suppose $u_1, \dots, u_n$ are linearly independent.
For any $v \in \text{span}\{u_1, \dots, u_n\}$, there exist **unique** $[\alpha_1, \alpha_2, \dots, \alpha_n]$ s.t.
$$v = \alpha_1 u_1 + \dots + \alpha_n u_n.$$

i.e., $v = Ax$ has exactly 1 solution, where $A = [u_1, \dots, u_n]$

Examples:

Independent: unique way to represent

$$\begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$a' \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b' \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Dependent: many ways to represent

$$\begin{bmatrix} 4 \\ 6 \end{bmatrix} = \textcircled{2} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \textcircled{0} \begin{bmatrix} 1 \\ 1.5 \end{bmatrix} = \underbrace{1}_{\cdot} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \underbrace{2}_{\cdot} \begin{bmatrix} 1 \\ 1.5 \end{bmatrix}$$

Full Row rank

Proposition 15.2: If $\text{rank}(A) = m$, i.e., full row rank, then $Ax = b$ has **at least one solution** ($\#$ of solutions must be 1 or ∞ ; **eliminate possibility of 0**)

From RREF perspective: no “zero rows” in RREF
 $\Rightarrow$ no equations like “0 = nonzero” $\Rightarrow$ at least one solution.

After re-ordering columns of RREF, we get:

Pivot columns free columns

pivot rows

free rows

not existing.

$$\begin{matrix} r \\ m-r \end{matrix} \left\{ \begin{matrix} \left[\begin{matrix} I & F \end{matrix} \right] \\ \left[\begin{matrix} 0 & 0 \end{matrix} \right] \end{matrix} \right\} \begin{bmatrix} x_p \\ x_F \end{bmatrix} = \begin{bmatrix} \hat{b} \\ c \end{bmatrix}$$

$\underbrace{\quad}_r \quad \underbrace{\quad}_{n-r}$

determine # of solutions

Relation to an earlier Result

Combine Prop. 15.1 and Prop. 15.2, we have

Corollary 15.2 If $\text{rank}(A) = m = n$, then $Ax=b$ has exactly one solution.
(corresponding to Case 1)

This is NOT a new result.

$\text{Rank}(A) = m = n$ means A is a full rank square matrix,
which is invertible.

So Corollary 15.2 can also be derived from invertibility conditions.

Part II Orthogonality

Column Form and Row Form

Linear independence:

a concept deeply related to the column form of $Ax=0$.

What can we get from the row form of $Ax=0$?

Motivation: Row Form

Motivation: Row form of $Ax=0$.

Row form:

$$\begin{bmatrix} \mathbf{a}_{(1)}^T \\ \mathbf{a}_{(2)}^T \\ \dots \\ \mathbf{a}_{(n)}^T \end{bmatrix} \mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} a_{(1)}^T x \\ \vdots \\ a_{(n)}^T x \end{bmatrix}$$

$$\Leftrightarrow a_{(i)}^T x = 0, \quad i=1, \dots, n.$$

Each equation is in the form of $u^T v = 0$.



Suppose $\text{angle}(u, v) = \theta$, then $\cos(\theta) = \frac{u^T v}{\|u\| \|v\|} = 0$,

So $\theta = 90^\circ$ is the right angle.

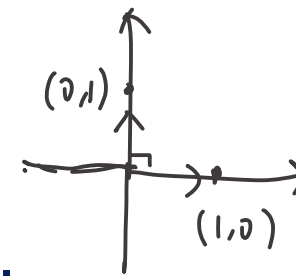
Orthogonality (正交)

Definition 15.2 (Orthogonal)

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if $\mathbf{u}^T \mathbf{v} = 0$. Denote $\mathbf{u} \perp \mathbf{v}$.

Examples:

Vectors $(1,0)$ and $(0,1)$ are orthogonal in $\mathbb{R}^2$.



$$3 \times 2 + (-4) \times 6 = 0$$

Vectors $[3, 2]^T$ and $[-4, 6]^T$ are orthogonal in $\mathbb{R}^2$.

Vectors $[2, -3, 1]^T$ and $[1, 1, 1]^T$ are orthogonal in $\mathbb{R}^3$.

$$2 \times 1 + (-3) \times 1 + 1 \times 1 = 0.$$

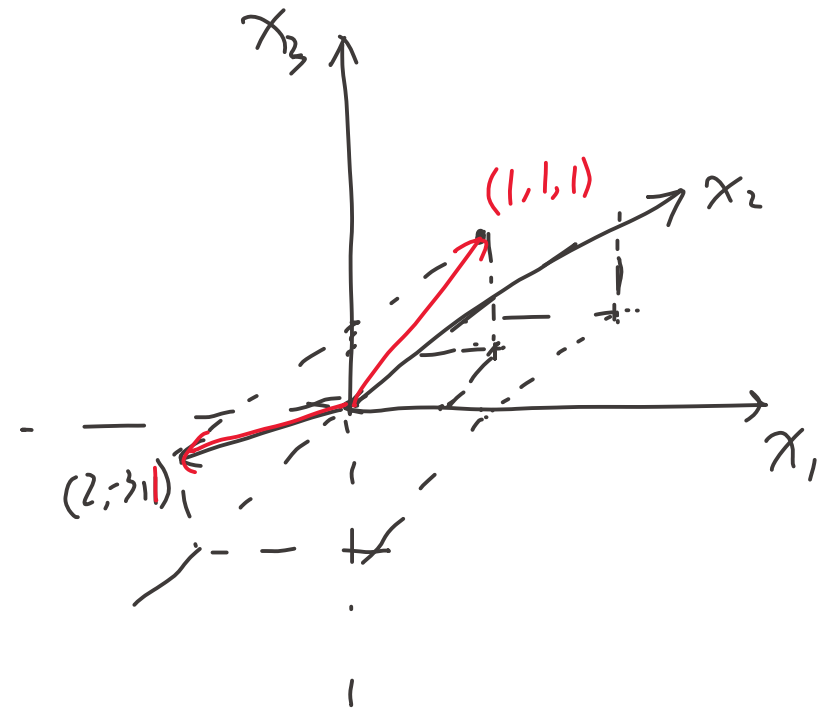
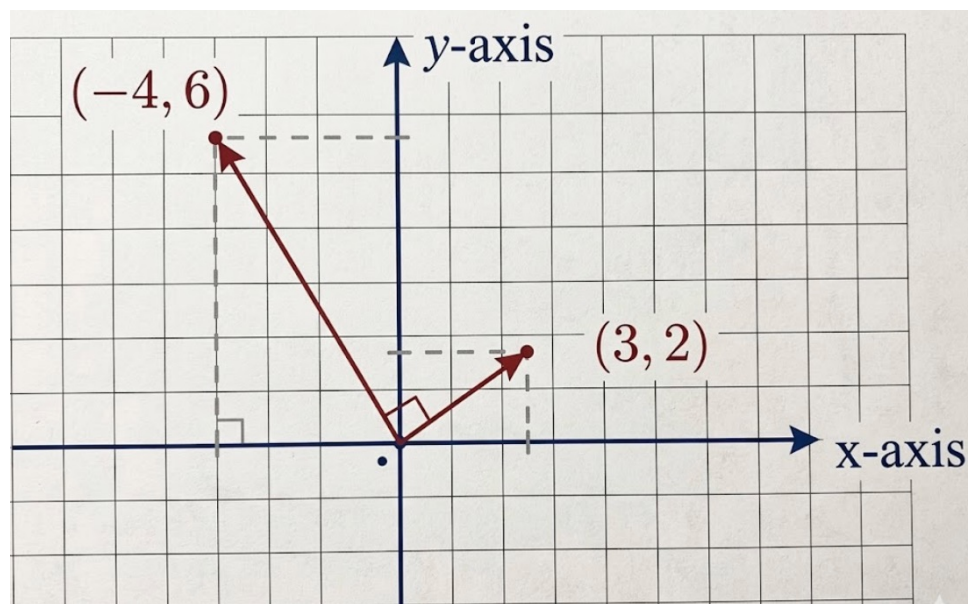
Examples of Orthogonality: Vector $\perp$ Vector

Examples:

Vectors $[3, 2]^T$ and $[-4, 6]^T$ are orthogonal in $\mathbb{R}^2$.

Vectors $[2, -3, 1]^T$ and $[1, 1, 1]^T$ are orthogonal in $\mathbb{R}^3$.

Geometry: (2D and 3D)



Recall: High School Math of Vector $\perp$ Space

High school math:

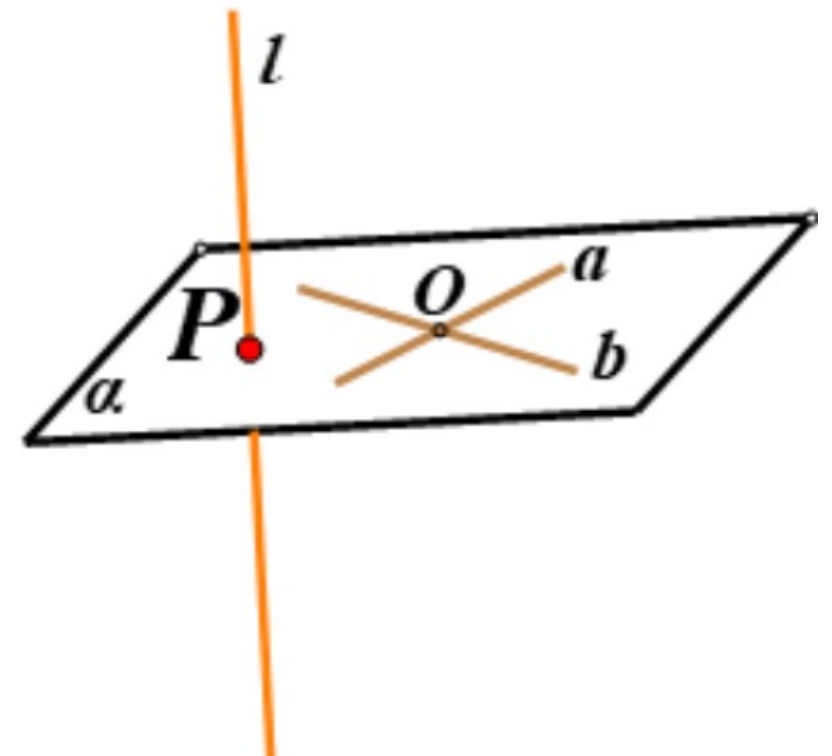
Def (line orthogonal to plane)

If a line is orthogonal to any line in the plane, then it

Criterion:

If a line $\perp$ two intersecting lines
then the line is orthogonal to the plane
spanned by the two lines.

$\Rightarrow$ The line is orthogonal to any
line on the plane.

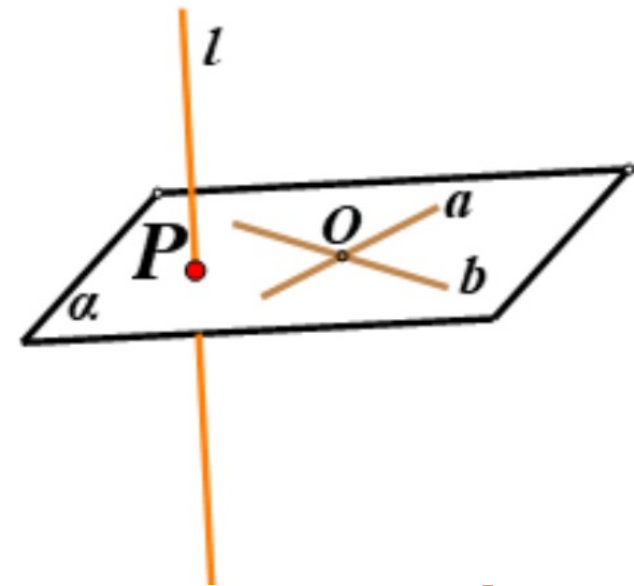


如果一条线垂直于另两条 相交直线, 那么垂直于两线张成的平面
(**线面垂直的判定定理**)

Recall: High School Math of Vector $\perp$ Space

High school geometry: vector $\perp$ 2D subspace

Extend: vector $\perp$ any-dim subspace



Definition (vector orthogonal to subspace)

We say a vector $\mathbf{u} \in \mathbb{R}^n$ is orthogonal to a subspace $\mathbf{V} \subseteq \mathbb{R}^n$, if $\mathbf{u} \perp \mathbf{v}, \forall \mathbf{v} \in \mathbf{V}$, denoted as $\mathbf{u} \perp \mathbf{V}$.

Eg 1: Floor and vertical flag are orthogonal

Claim If $\mathbf{u} \perp \mathbf{v}_j, j = 1, \dots, n$, then $\mathbf{u} \perp \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$

$$\begin{aligned} \mathbf{u}^T (\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) &= \alpha_1 \mathbf{u}^T \mathbf{v}_1 + \dots + \alpha_n \mathbf{u}^T \mathbf{v}_n \\ \Rightarrow \mathbf{u} \perp \sum_{i=1}^n \alpha_i \mathbf{v}_i, \forall \alpha_i \in \mathbb{R} &= 0. \end{aligned}$$

Extension: Space $\perp$ Space

$\text{Span}\{v\} \perp \text{span}\{u_1, u_2\}$
one vec two vec

If $v \perp \text{span}\{u_1, u_2\}$,
Any $\alpha v \perp \text{span}\{u_1, u_2\}$.

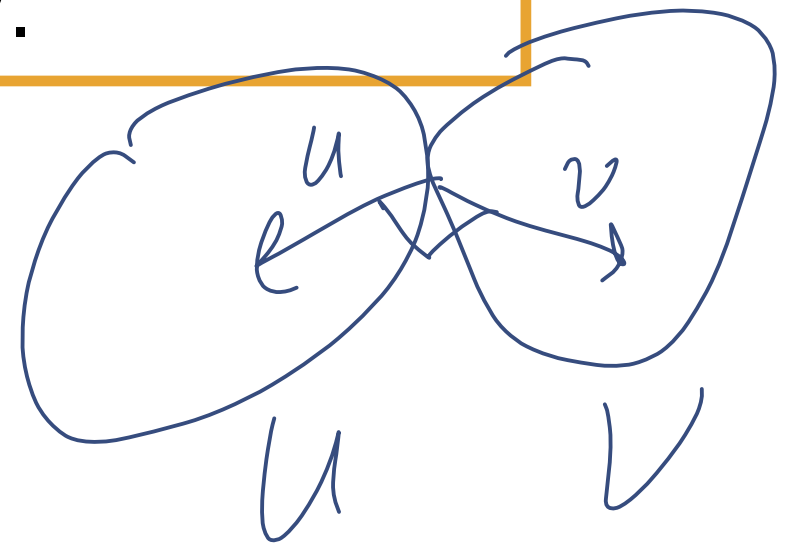
Extend: we say $\text{Span}\{v_1, v_2\} \perp \text{span}\{u_1, u_2\}$
iff $\alpha_1 v_1 + \alpha_2 v_2 \perp \beta_1 u_1 + \beta_2 u_2$.

Definition (Orthogonal subspaces)

We say two subspaces $\mathbf{U}, \mathbf{V} \subseteq \mathbb{R}^n$ are orthogonal if $\mathbf{u} \perp \mathbf{v}, \forall \mathbf{u} \in \mathbf{U}, \mathbf{v} \in \mathbf{V}$. Denoted as $\mathbf{U} \perp \mathbf{V}$.

Eg: In $\mathbb{R}^5$, $\text{span}\{e_1, e_2\} \perp \text{span}\{e_3, e_4\}$.

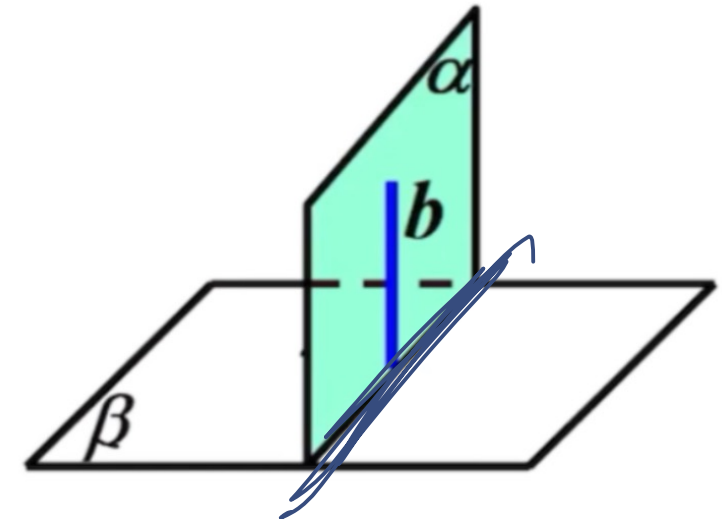
$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \perp \begin{bmatrix} 0 \\ 0 \\ \beta_1 \\ \beta_2 \\ 0 \end{bmatrix}$$



Pay Attention: Different with High-School Math

High school math: orthogonal planes mean

Two planes are orthogonal if one plane contains a line perpendicular to the other plane.



中文翻译 (供参考):

如果一个平面包含一条直线垂直于另一个平面，那么这两个平面互相垂直。

Note: This kind of orthogonality is different in linear algebra—require every vector in one subspace to be orthogonal to every vector in the other."

Non-example: Floor and wall are not orthogonal.

i.e., in 3D space, $\text{span}\{e_1, e_2\}$ and $\text{span}\{e_1, e_3\}$ are NOT orthogonal



Set of All Orthogonal Vectors?

In 3D space, what vectors are orthogonal to $e_3 = [0,0,1]^T$?

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \perp \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \dots$$

$$\left\{ \begin{bmatrix} \alpha \\ \beta \\ 0 \end{bmatrix} \right\}$$

Collect all vectors that are orthogonal to $e_3 = [0,0,1]^T$, form a plane space.

Orthogonal Complement of Subspace

There can be many vectors that are orthogonal to subspace U . Collect all of them, we obtain the orthogonal complement.

Definition (Orthogonal complement) 正交补 (空间).

For any subspace $\mathbf{V} \subseteq \mathbb{R}^n$, the set of vectors that are orthogonal to $\mathbf{V}$
$$\{\mathbf{u} \in \mathbb{R}^n \mid \mathbf{u} \perp \mathbf{v}, \forall \mathbf{v} \in \mathbf{V}\}.$$

is called the orthogonal complement of $\mathbf{V}$, denoted as $\mathbf{U} = \mathbf{V}^\perp$.
(reads "V perp")

Remark: A subspace V has a **unique** orthogonal complement! (in $\mathbb{R}^n$).

Eg 1: Floor = (vertical flag) $^\perp$,

Or in 3D space, $\text{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \text{span}(\{\mathbf{e}_3\})^\perp$
$$\left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \right\} \quad \left\{ \begin{bmatrix} 0 \\ 0 \\ y \end{bmatrix} \right\}.$$

More Examples and Non-example

Eg 2: in 2D, $x\text{-axis} = (y\text{-axis})^\perp$,

Or $\text{span}\{e_1\} = \text{span}\{e_2\}^\perp$

Others:

In $\mathbb{R}^3$, $(\overbrace{\text{span}\{e_1, e_2\}}^{\dim 2})^\perp = \overbrace{\text{span}\{e_3\}}^{\dim 1}$.

In $\mathbb{R}^5$, $(\text{span}\{e_1, e_2\})^\perp = \text{span}\{e_3, e_4, e_5\}$

In $\mathbb{R}^3$, $(\text{span}\{(1,1,1)\})^\perp = \{\mathbf{x} \in \mathbb{R}^3 \mid \mathbf{x} \cdot (1,1,1) = 0\} = \{\mathbf{x} \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$

Non-example: One line on the floor $\neq$ (vertical flag) $^\perp$

Or in $\mathbb{R}^3$, $\underbrace{\text{span}\{e_1\}}_{\dim 1} \neq (\underbrace{\text{span}\{e_2\}}_{\dim 1})^\perp$.

$\text{span}\{e_1\} \perp \text{span}\{e_2\}$

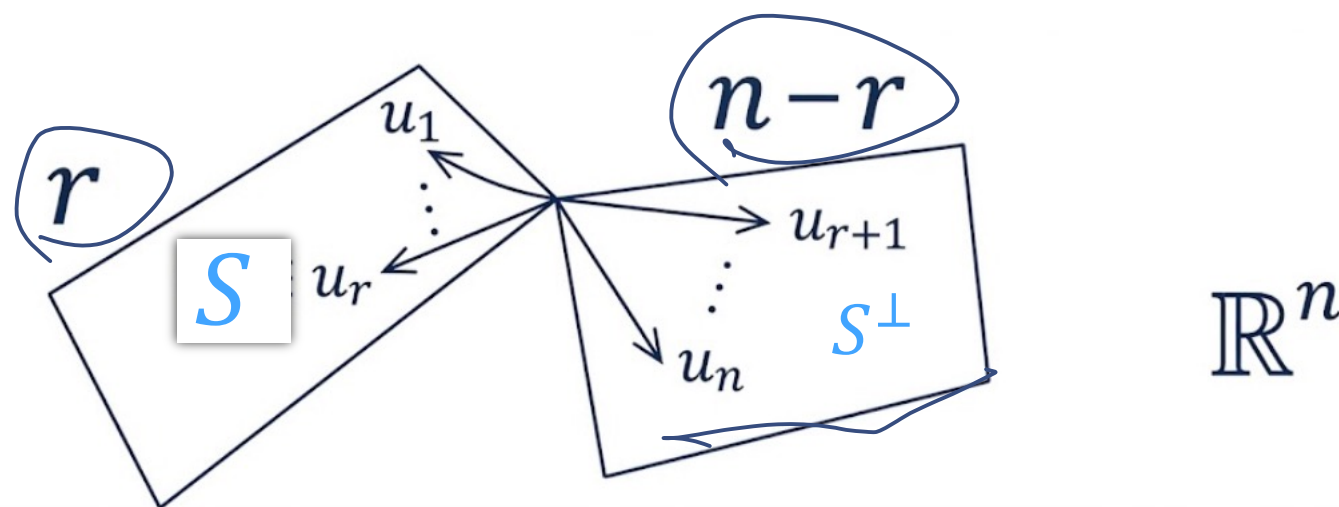
Dimension of Orthogonal Complement

Theorem 15.2 (dim of V perp)

Suppose S is a subspace of $\mathbb{R}^n$. Then the following hold:

- i) $S^\perp$ is a subspace;
- ii) $\dim(S) + \dim(S^\perp) = n$.
- iii) If $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is a basis of S , and $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_n\}$ is a basis of $S^\perp$, then $\{\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_n\}$ is a basis of $\mathbb{R}^n$.

Remark: $(S^\perp)^\perp = S$.



Proof: [skip]. can use rank-nullity theorem, or Gauss-Schmit process].

Decomposition into Two Parts

Corollary:

Suppose $S = T^\perp \subseteq \mathbb{R}^n$. Then any $u \in \mathbb{R}^n$ can be written as $u = u_S + u_T$, where $u_S \in S, u_T \in T$.

$$\mathbf{u} = \underbrace{\alpha_1 \mathbf{v}_1 + \cdots + \alpha_r \mathbf{v}_r}_{=\mathbf{u}_S \in S} + \underbrace{\alpha_{r+1} \mathbf{v}_{r+1} + \cdots + \alpha_n \mathbf{v}_n}_{=\mathbf{u}_T \in T}$$

$$u_s \perp u_T$$

where $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ forms a basis for S , and $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$ forms a basis for T . $T = S^\perp$

e.g., $n = 3$, $S = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$, $T = \text{span}(\mathbf{e}_2)$, $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}}_{\in S} + \underbrace{\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}}_{\in T}$

$$\text{e.g., } n = 3, S = \text{span}\{(1, 1, 1)\}, T = S^\perp = \{\mathbf{x} \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\},$$

$$\begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}}_{\in S} + \underbrace{\begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}}_{\in T}$$

elements sum to 0

Part III Four Fundamental Subspaces

Strang's book Sec. 3.5

Important Relation Hidden

$$\begin{aligned}(\text{span}\{(1, 1, 1)\})^\perp &= \{\mathbf{x} \in \mathbb{R}^3 \mid \mathbf{x} \perp \alpha(1, 1, 1) \forall \alpha \in \mathbb{R}\} \\&= \{\mathbf{x} \in \mathbb{R}^3 \mid \alpha(x_1 + x_2 + x_3) = 0 \forall \alpha \in \mathbb{R}\} \\&= \{\mathbf{x} \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}.\end{aligned}$$

Let $A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$. Then

Q: What are the two sets related to $A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$?

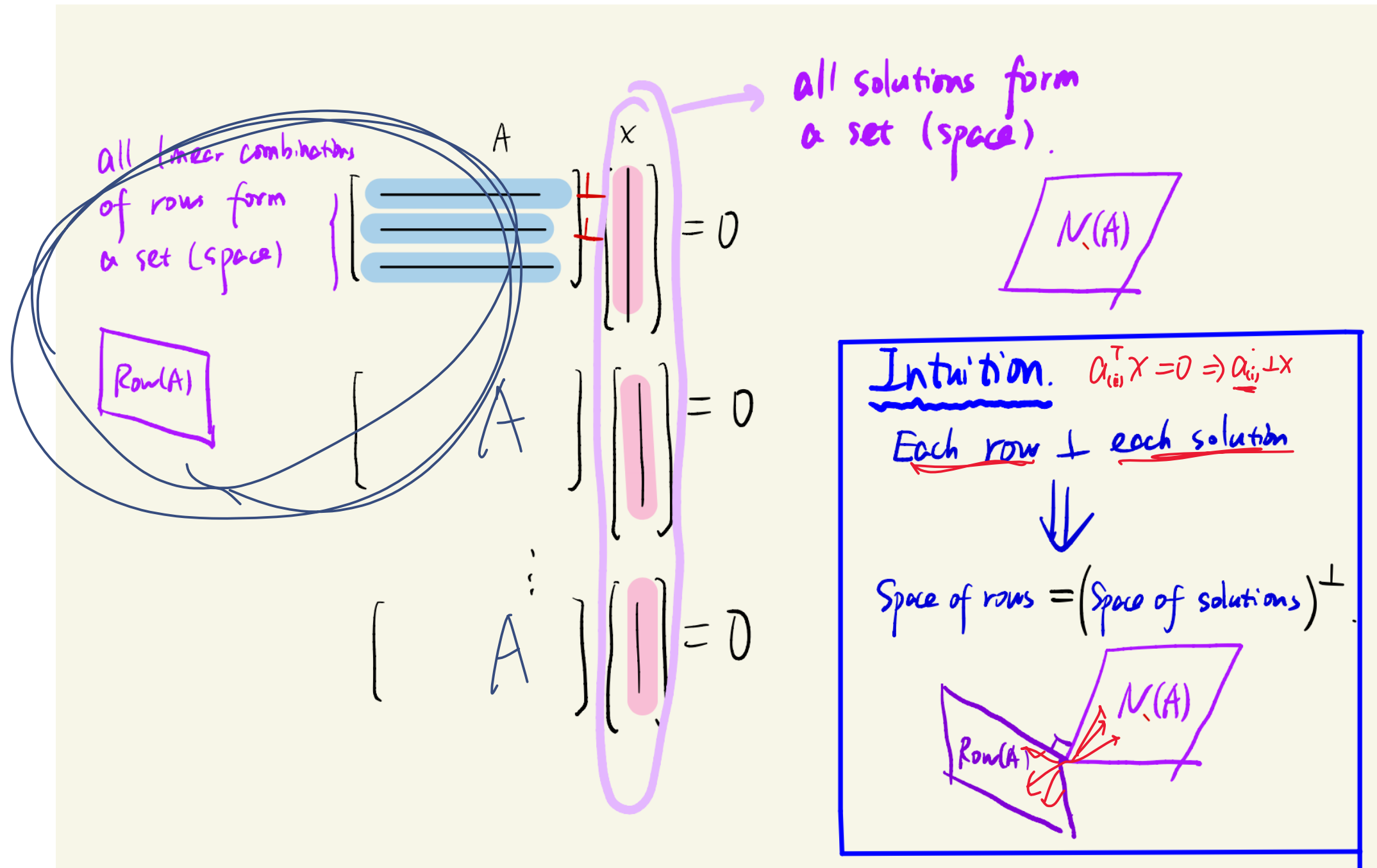
Left set: $(\text{span}\{(1, 1, 1)\})^\perp = \text{Row}(A)^\perp$

Right set: $\{\mathbf{x} \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\} = \mathcal{N}(A)$

$$\boxed{\text{Row}(A)^\perp = \mathcal{N}(A)}$$

Intuition: Row Space v.s. Null Space

Proposition: $N(A) = \text{Row}(A)^\perp$



Relation of Row Space & Null Space

Proposition: $N(A) = \text{Row}(A)^\perp$

Proof:

$$A = \begin{bmatrix} a_{(1)}^\top \\ \vdots \\ a_{(m)}^\top \end{bmatrix}$$

$\mathcal{N}(A)$ = Solution set of $A\mathbf{x} = \mathbf{0}$

$$= \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_{(1)}^\top \mathbf{x} = \cdots = \mathbf{a}_{(m)}^\top \mathbf{x} = 0 \right\}$$

$$= \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \perp a_{(1)}, \dots, \mathbf{x} \perp a_{(m)} \right\}$$

$$= \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \perp \text{span}\{a_{(1)}, \dots, a_{(m)}\} \right\}$$

$$= \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \perp \text{Row}(A) \right\}$$

$$= \text{Row}(A)^\perp.$$

[Definition]

[write as orthogonal]

span [linear combination]

[Definition of $\text{Row}(A)^\perp$]

[Def of $(\text{set})^\perp$]

Hint: use the following result.

Claim If $\mathbf{u} \perp \mathbf{v}_j, j = 1, \dots, n$, then $\mathbf{u} \perp \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$

Left Null Space

Definition 15.3 (left null space)

The left null space of a matrix A is defined as $N(A^\top)$.

Think: What is this space?

Hint: using linear equations.

$$\begin{aligned}\mathcal{N}(A^\top) &= \{\mathbf{y} \in \mathbb{R}^m \mid A^\top \mathbf{y} = \mathbf{0}\} \\ &= \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{y}^\top A = \mathbf{0}\}.\end{aligned}$$

e.g. matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$\begin{aligned}\mathbf{y}^\top A = \mathbf{0} &\iff \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix} \\ &\iff \begin{cases} y_1 + 3y_2 = 0, \\ 2y_1 + 4y_2 = 0. \end{cases}\end{aligned}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Compare

$$\begin{aligned}\mathcal{N}(A) &= \{\mathbf{x} \in \mathbb{R}^2 \mid A\mathbf{x} = \mathbf{0}\} \\ A\mathbf{x} = \mathbf{0} &\iff \begin{cases} x_1 + 2x_2 = 0, \\ 3x_1 + 4x_2 = 0. \end{cases}\end{aligned}$$

Left Null Space and Column Space

We learned:

$$\mathcal{N}(K) = \text{Row}(K)^\perp, \quad \text{for any matrix } K$$

$$\begin{array}{c} K = A \\ \downarrow \\ \mathcal{N}(A) = \text{Row}(A)^\perp \end{array}$$

$$\begin{array}{c} K = \underline{A^\top} \\ \downarrow \\ \mathcal{N}(A^\top) = \boxed{\text{Row}(A^\top)}^\perp \\ = \boxed{C(A)}^\perp \end{array}$$

The null space of $A^\top$ is the orthogonal complement of the column space of A .

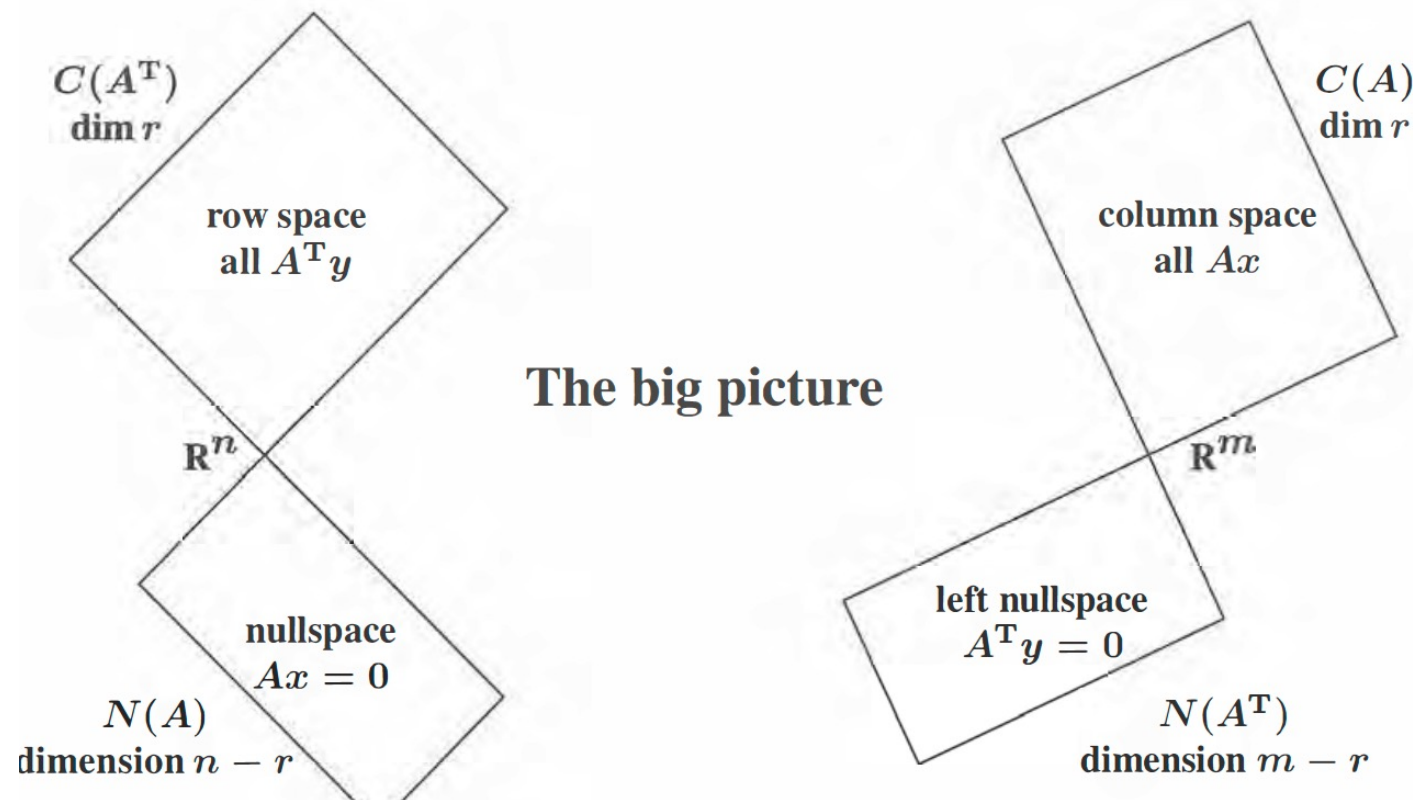
Fundamental Theorem of Linear Algebra

Theorem 15.3 (Fundamental Theorem of Linear Algebra)

Suppose $A \in \mathbb{R}^{m \times n}$. Then:

(1) $N(A) = \text{Row}(A)^\perp$, $\dim(N(A)) = n - \text{rank}(A)$.

(2) $N(A^\top) = C(A)^\perp$, $\dim(N(A^\top)) = m - \text{rank}(A)$.



Theorem (rank-nullity theorem)

Suppose A has n columns. Define the nullity of A as $\dim(N(A))$.

$$\text{rank}(A) + \dim(N(A)) = n.$$

$$\dim(N(A)) = n - r,$$

Exercise: Dim of Each Subspace

Recall: m equations
(m rows)

n variables
(n columns)



$C(A)$, Subspaces of $\mathbb{R}^m$

$C(A^T)$, Subspaces of $\mathbb{R}^n$

row vector
size n

thus $\text{Row}(A) \subseteq \mathbb{R}^n$

Column space dim r

=

Row space dim r

Ortho-complement

sum = m

Ortho-complement

sum = n

Left null space dim $m-r$

$N(A^T)$, Subspaces of $\mathbb{R}^m$

null space dim $n-r$

$N(A)$, Subspaces of $\mathbb{R}^n$

set of x .

each $x \in \mathbb{R}^n$

thus $N(A) \subseteq \mathbb{R}^n$

$$\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} a_{1j} \\ \vdots \\ \vdots \end{bmatrix} = 0.$$

$x \in \mathbb{R}^n$

m rows
Col vector
size m
 $C(A) \subseteq \mathbb{R}^m$

Summary of Solving Rectangular Linear Systems

What have we learned, for solving $Ax=b$?

1) For any linear system, can transform to RREF.

Lec 11

2) From RREF, can write the solution set as $x_p + N(A)$,
where $N(A)$ is the span of $(n-r)$ vectors

Lec 13

Lec 15

A new view for solving $Ax=0$:

It is just finding

orthogonal complement of $\text{Row}(A)$.

Lec 15



Summary Today (write Your Own)

One sentence summary:

Detailed summary:

Summary Today (of Instructor)

One sentence summary:

Four Fundamental Subspaces

Detailed summary:

1. Rank and Solutions

--m,n, r determine # of solutions

—Full row rank $\implies$ at least 1 solution

--Full column rank $\implies$ at most 1 solution

(last lecture: when $m < n$, $Ax=0$ has infinitely many solutions)

2. Orthogonality

—Orthogonal complement $V^\perp$ has dimension $n - \dim(V)$

3. Four fundamental subspaces

—null space = (row space) $^\perp$; left null space = (column space) $^\perp$.

—dim: $m - r, n - r$

--Rank-Nullity theorem: Nullity (dimension of null space) + rank = n.

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